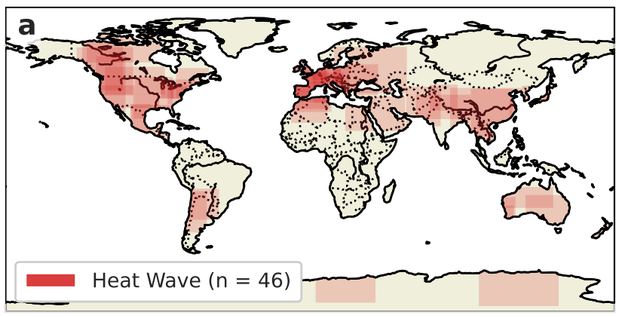
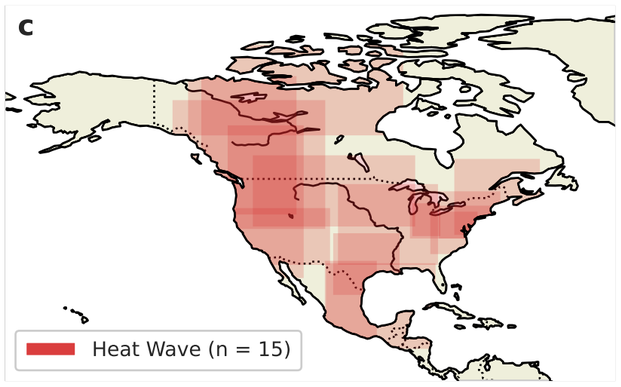
All Heat Waves



b

	RMSE					MAE of Max Temp					MAE of Highest Min Temp				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	2.7	2.9	3.3	3.8	4.5	1.1	1.3	1.4	1.9	6.4	1.2	1.2	1.3	1.5	2.2
GraphCast	2.6	2.7	3.0	3.5	4.4	1.2	1.3	1.5	2.0	3.4	1.1	1.2	1.3	1.3	2.2
Pangu	2.6	2.8	3.1	3.7	4.5	1.0	1.1	1.5	2.2	3.7	0.9	1.0	1.2	1.6	2.6
AIFS	2.4	2.5	2.8	3.3	4.2	0.9	1.1	1.3	1.8	3.1	0.9	0.9	1.1	1.3	2.0

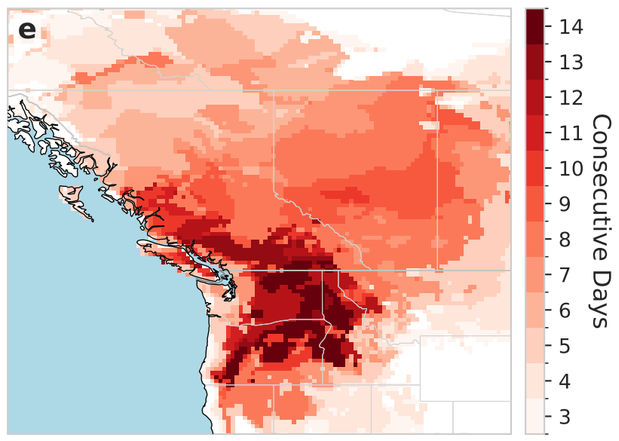
North America Heat Waves



d

	RMSE					MAE of Max Temp					MAE of Highest Min Temp				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	2.6	2.8	3.2	3.6	4.3	0.8	1.1	1.0	1.4	5.9	0.8	0.8	0.9	1.0	1.3
GraphCast	2.4	2.6	2.9	3.4	4.4	1.0	1.1	1.3	1.4	3.4	0.8	0.8	0.8	0.7	2.0
Pangu	2.4	2.6	2.9	3.5	4.4	0.7	0.8	1.0	1.4	3.2	0.6	0.7	0.8	1.0	2.0
AIFS	2.3	2.4	2.6	3.1	4.1	0.7	0.9	0.9	1.3	2.9	0.6	0.7	0.8	0.9	1.6

Consecutive Heat Wave Days
2021-06-20 to 2021-07-03

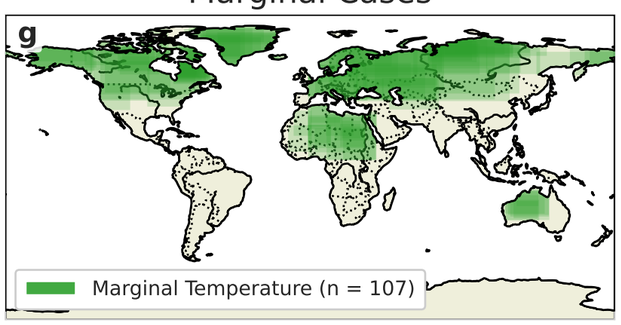


f

	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	2.8	3.0	3.7	4.4	5.8	0.9	0.5	0.9	0.5	11.7	0.6	0.8	0.5	1.0	1.3
GraphCast	2.7	2.8	3.3	4.1	5.7	1.1	1.1	1.9	0.7	8.4	0.1	0.1	0.1	0.1	1.8
Pangu	2.6	2.9	3.5	4.3	6.1	0.5	0.7	0.9	0.8	7.8	0.5	0.6	0.9	1.6	2.3
AIFS	2.5	2.6	3.0	3.7	4.9	0.7	0.8	1.5	1.3	4.2	0.7	0.9	1.0	1.5	1.2

Lead Time [Days] | Lead Time [Days] | Lead Time [Days]

Marginal Cases



h

RMSE

	1	3	5	7	10
HRES IFS	3.0	3.3	3.8	4.5	5.3
GraphCast	2.8	3.0	3.5	4.1	5.1
Pangu	2.7	3.1	3.6	4.4	5.6
AIFS	2.6	2.8	3.3	3.9	5.0

Lead Time [Days]

